Laser-Induced Graphene: A Scalable Platform for Low-Cost Biosensors in Early Diabetes Detection

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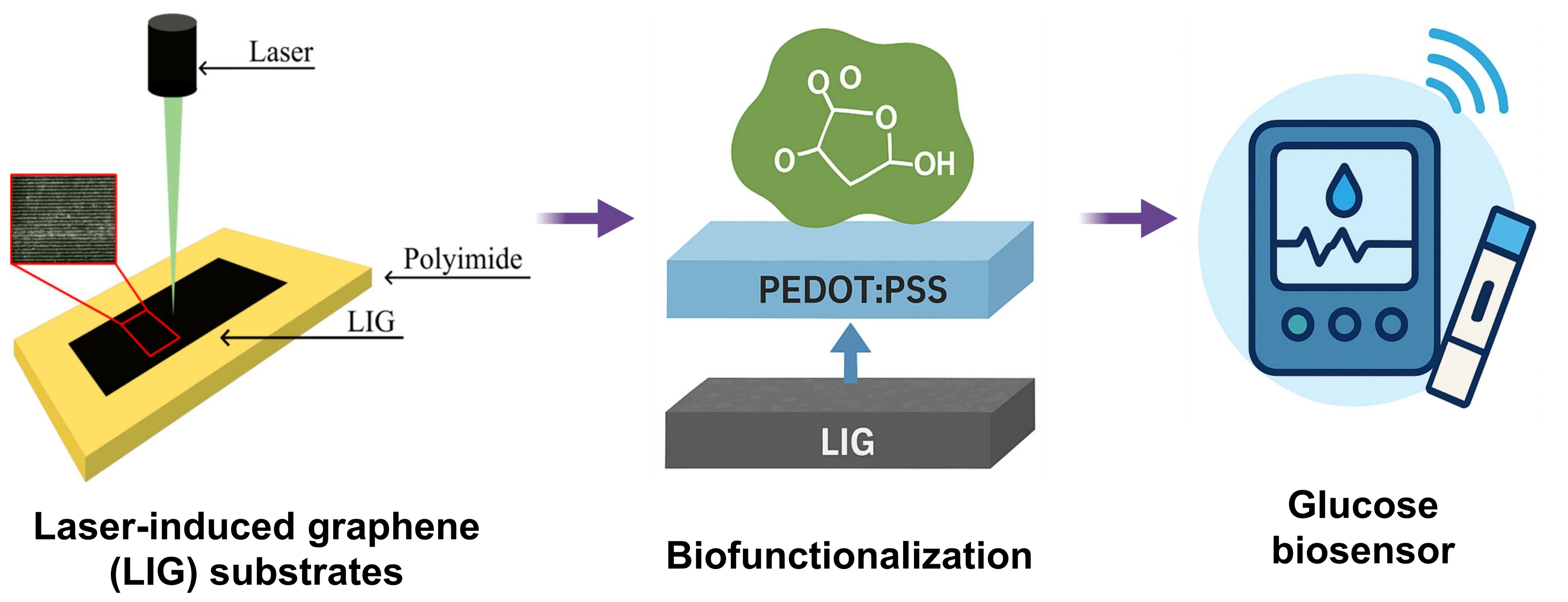
1 Introduction

Access to healthcare remains limited in rural and low-resource regions, where chronic diseases such as diabetes are often diagnosed late due to the lack of affordable diagnostic tools. Bridging this technological and health gap requires the development of low-cost, portable biosensing platforms adaptable to these settings. Graphene (G) stands out for its exceptional conductivity, flexibility, and biocompatibility, making it a promising material for biomedical applications. Laser-Induced Graphene (LIG) offers a scalable and sustainable approach to produce graphene-like conductive films by laser irradiation of polyimide (PI). In this work, LIG substrates were synthesized and characterized as precursors for future biosensor development. Structural, morphological, and wettability analyses confirmed the successful formation of conductive, porous graphene-like networks suitable for subsequent biofunctionalization.

This research establishes a foundation for developing affordable and adaptable biosensing platforms designed to enhance early disease detection and promote health equity in underserved populations.

2 Objectives

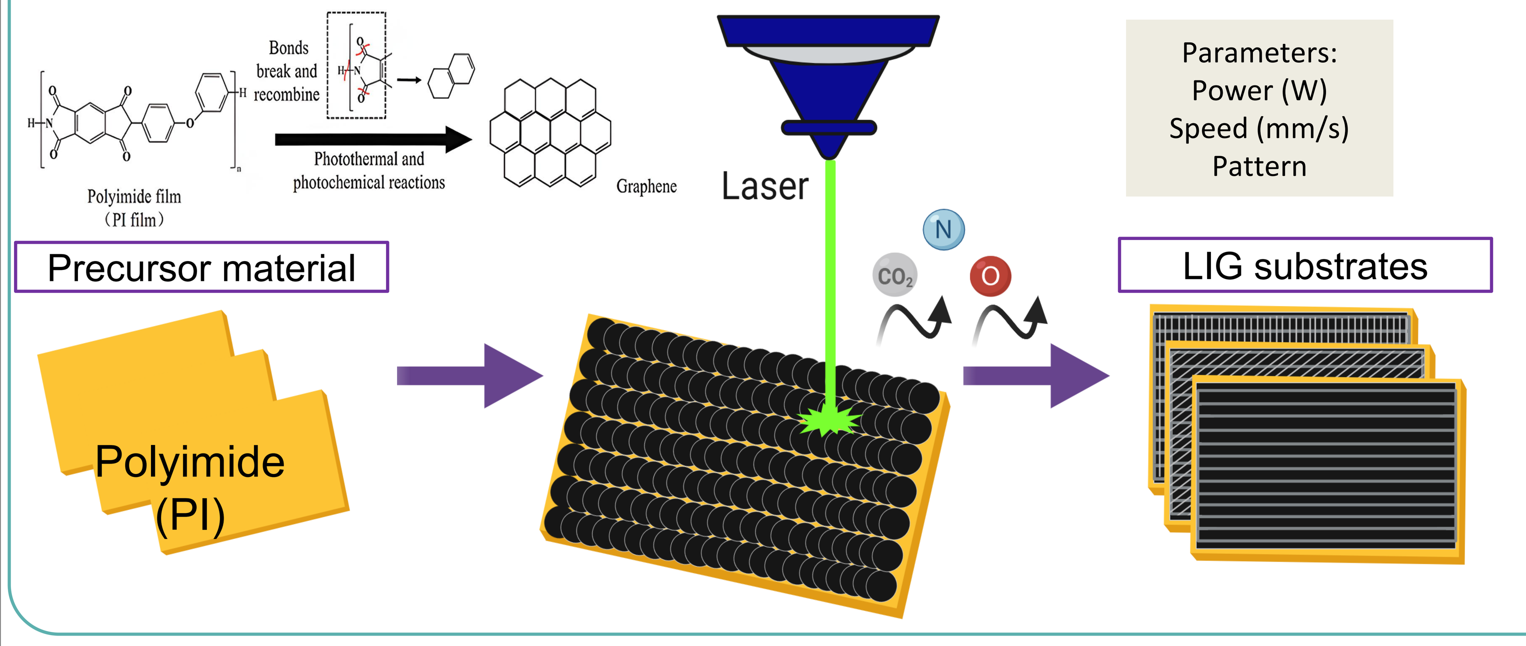
To develop portable and low-cost biosensors based on laser-induced graphene (LIG) for early detection of chronic disease biomarkers in low-resource and medically underserved communities worldwide.



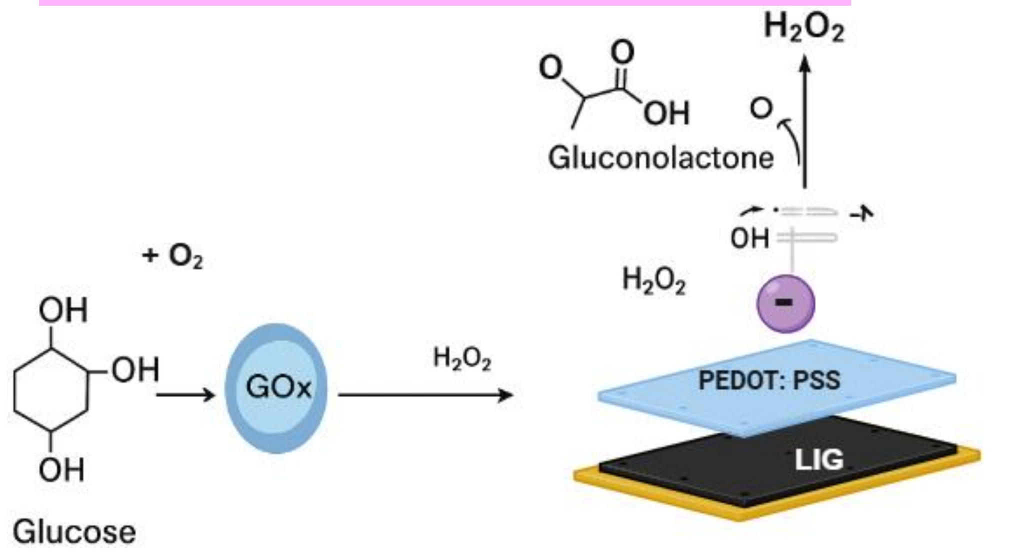
3 Materials and methods

Laser-Induced Graphene (LIG) fabrication

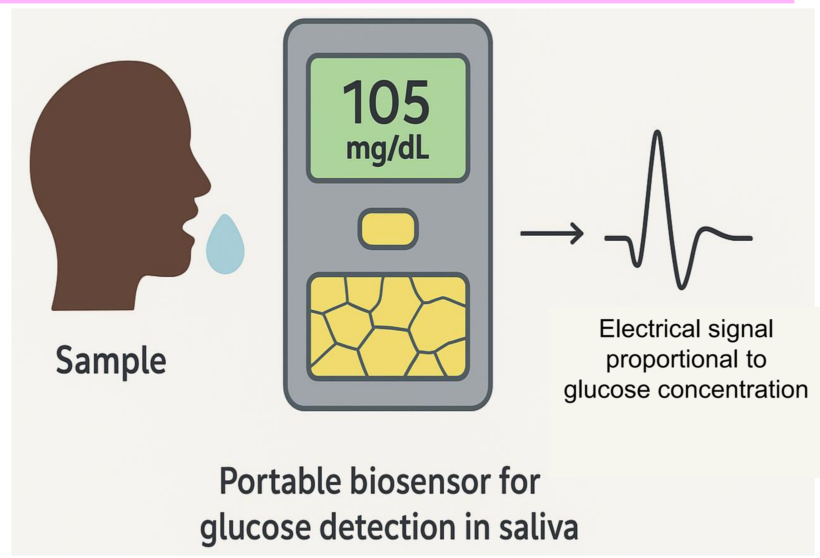
For the laser Induction process, we use a pulsed light laser, an instrument of precision that allows us to manipulate matter at the nanoscale.



Surface functionalization



Glucose biosensor

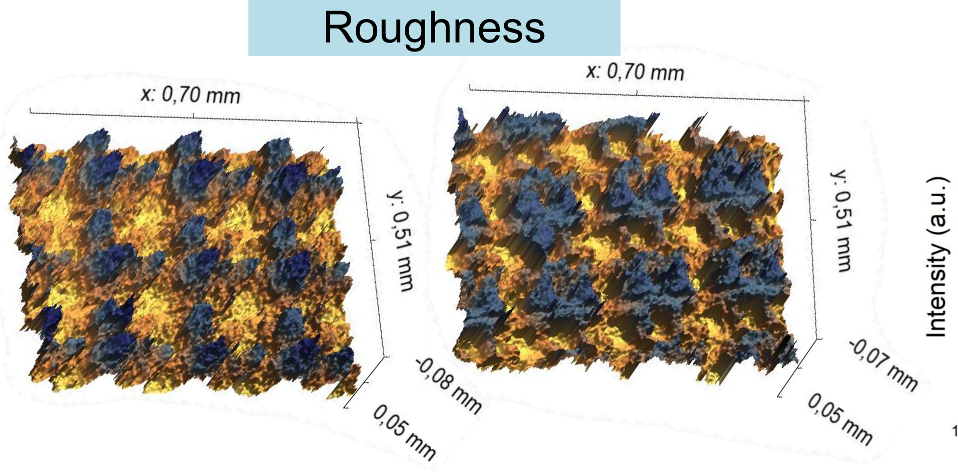


To transform the LIG into a functional biosensor, a thin conductive coating (PEDOT:PSS) was added to improve stability and electron flow. Then, the enzyme glucose oxidase (GOx) was attached to the surface, it acts as the biological "detector" that reacts with glucose. This process turns the LIG material into an active sensing platform capable of recognizing glucose in biological samples such as blood or saliva.

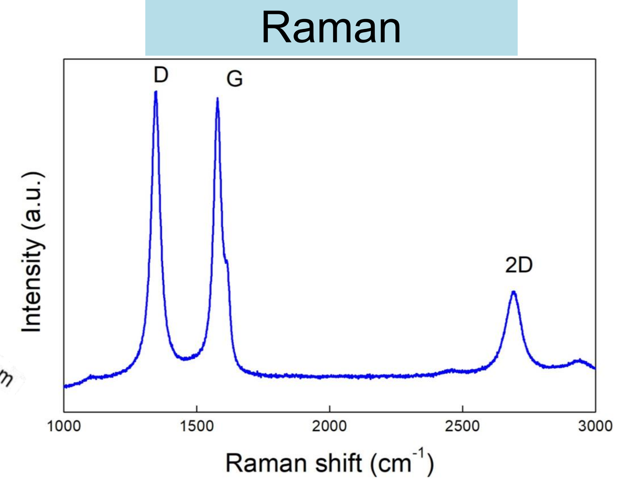
4 Results

1. Physicochemical characterization

Roughness

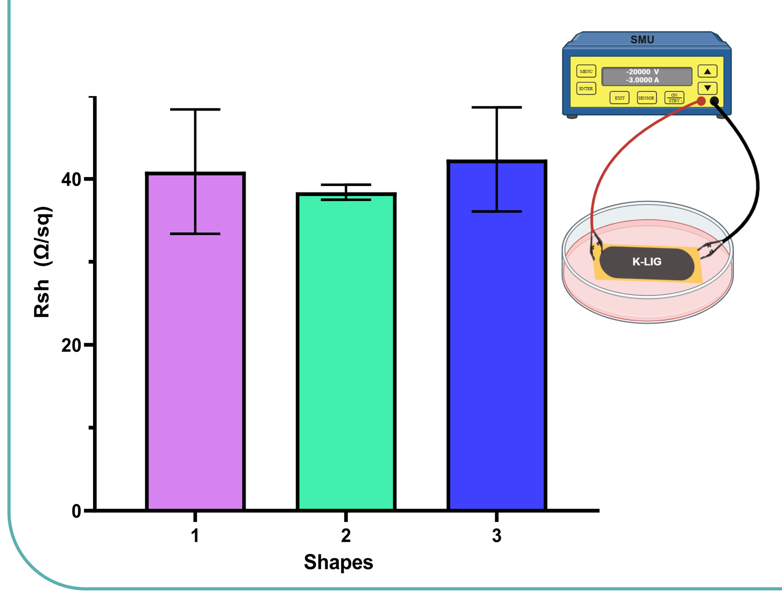


Raman

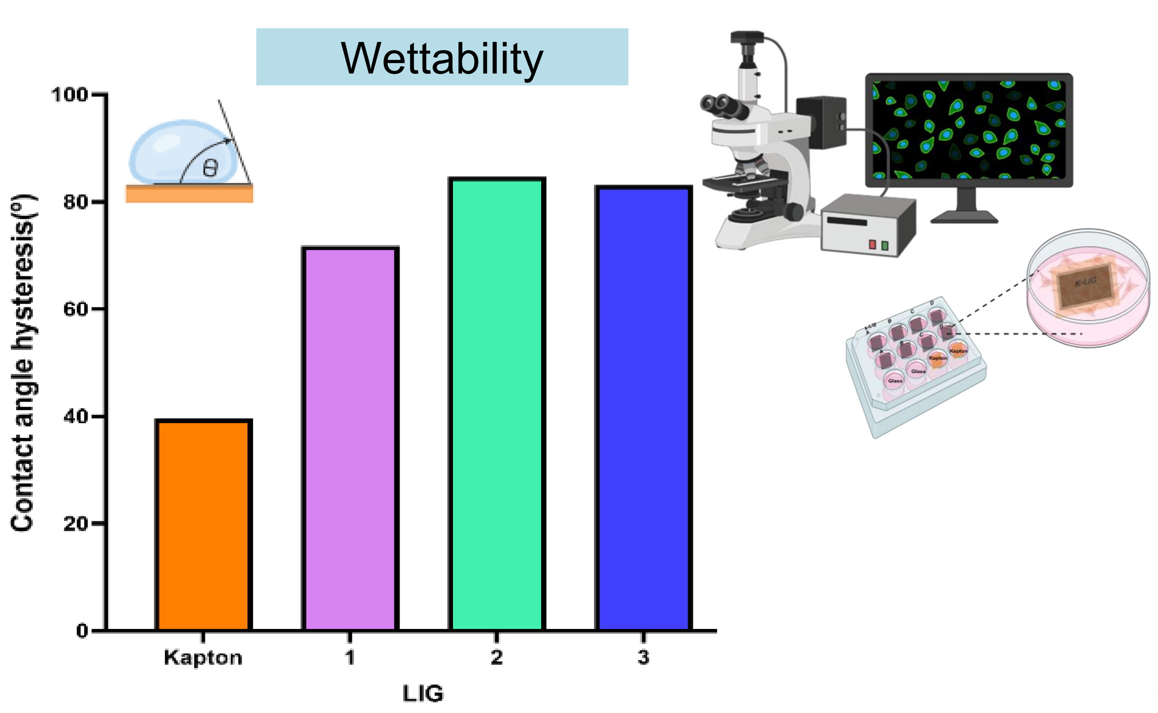


LIG substrates exhibited a 3D porous carbon network with high electrical conductivity, suitable for biosensing interface design.

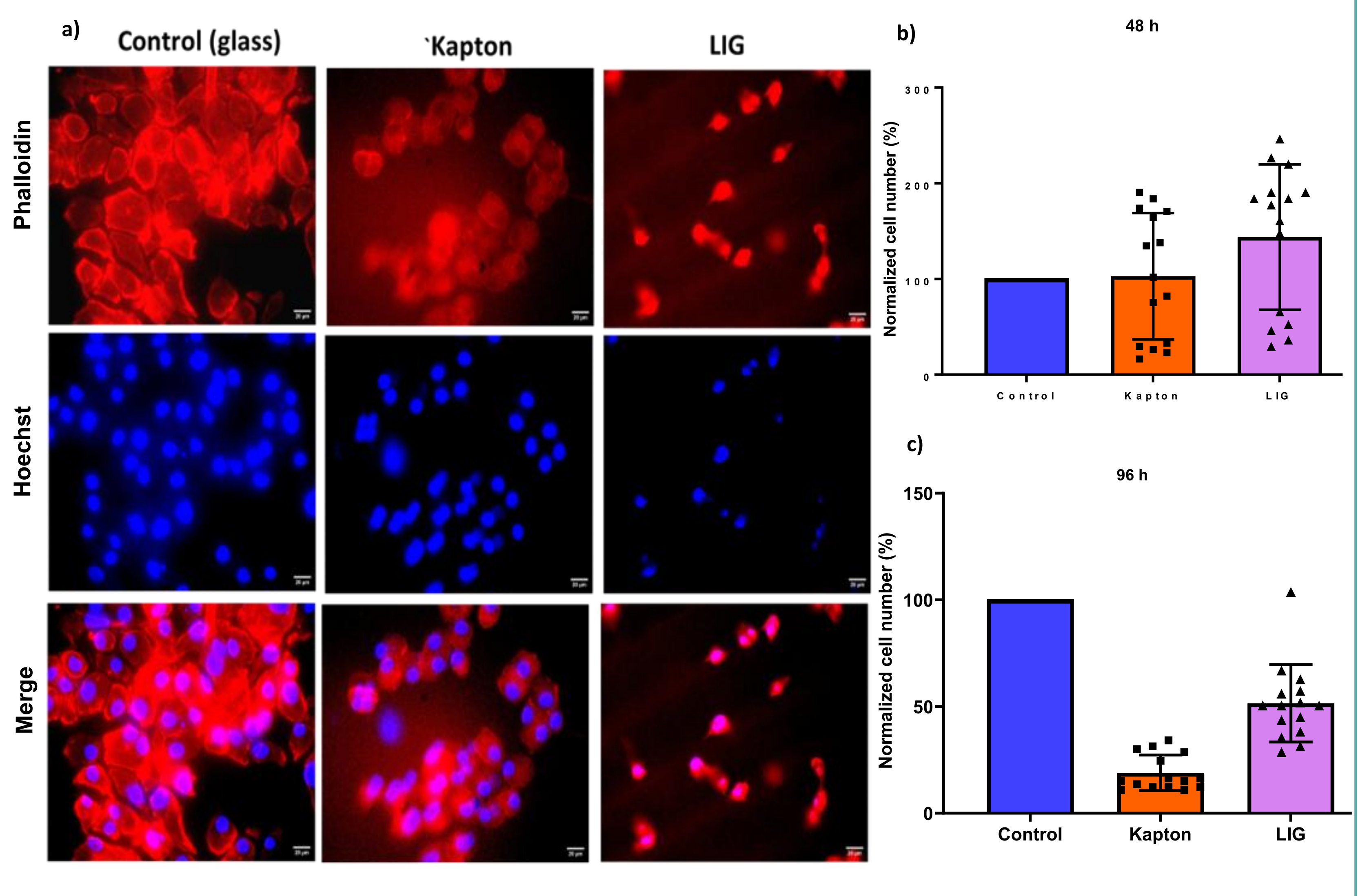
Electrical conductivity



Wettability



2. Biocompatibility Assessment of LIG Substrates



Biological bicompatibility assay. a) Visualization under fluorescence microscopy of the interaction of the U-87 MG cell line seeded on glass (control), Kapton**, c)** K-LIG. Hoechst (blue) stains the nuclei, and phalloidin (red) stains the actin cytoskeleton after 96 hours of cell culture. b) Quantitative analysis of cell density (% relative to Control) after 48 and c) 96 hours of culture on substrates).****p < 0.0001 vs. Control.

4 Conclusions

- 🌿 LIG substrates were successfully fabricated by laser scribing of polyimide (kapton).
- 🔬 Physicochemical analyses (Raman, XPS, contact angle) confirmed a porous, graphene-like conductive structure.
- ⚡ Laser parameter tuning enabled control over conductivity, roughness, and wettability, enhancing substrate versatility.
- 💉 Preliminary biological assays confirmed good biocompatibility, supporting their suitability for biosensing interfaces.

Future work: enzymatic functionalization with glucose oxidase (GOx) to enable selective glucose detection and development of portable, low-cost biosensors for use in resource-limited environments.

Our work moves toward accessible biosensing solutions that enable early diagnosis and support health equity worldwide.

5 References

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